SIMATS ENGINEERING

SIMATS
Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

CO2 - ASSESSMENT 1

REPORT ANALYSIS

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Course Faculty	**:**	**Dr.K.Anbazhagan**
Course Name	**:**	**Ethical Hacking**
Course Code	**:**	**ITA1406**
Slot	**:**	**Slot B**
Total Marks	**:**	**100**

CO2 :

Foot printing Tools: Conducting Competitive Intelligence, DNS Zone Transfers, Introduction to Social Engineering: Social Engineering Attacks and Countermeasures. (BT3)

Code of Conduct:

I, P. Venkata Sai Pradeep Reddy (Register Number: 192311364) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: ***PVS***

Answer the Following

Steps followed to do Domain & Email Footprinting using theHarvester:

Step 1: Install Python

- Open https://www.python.org/downloads/
- Download the latest Python version.
- Run the installer.
- Tick Add Python to PATH
- Click Install Now

Step 2: Verify Python Installation

- Press Windows + R
- Type cmd and press Enter
- Run: [python -m pip install --upgrade pip] in cmd

Step 4: Install theHarvester

- Run: [pip install theHarvester] in cmd
- Run: [python -m pip install theHarvester] in cmd

- Wait until the installation finishes.

Step 5: Verify theHarvester

- Run: [theHarvester --help] in cmd

Step 6: Run theHarvester

- Run: [theHarvester -d saveetha.com -b all] in cmd

Step 7: Wait for the Results

- After a few minutes, theHarvester displays information such as: Email Addresses, Subdomains, Host Names and IP Addresses.

Installation Note (what was actually done)

Running pip install theHarvester installed version 0.0.1 of a package of that name from PyPI, but the command theHarvester was then "not recognized", so the tool could not be verified that way. The tool was therefore installed from its official GitHub repository instead: (1) git clone https://github.com/laramies/theHarvester.git, (2) cd theHarvester, (3) pip install uv, (4) uv sync. It was verified with uv run theHarvester -h, which displayed the banner "theHarvester 5.0.0" and the full list of options. In the commands below, "uv run theHarvester" takes the place of "theHarvester".

Two scans were run against the domain saveetha.com:

- **Scan 1:** uv run theHarvester -d saveetha.com -b all -f saveetha_results (Scan 1: all passive sources; results saved to saveetha_results.json / .xml / .jsonl).
- **Scan 2:** uv run theHarvester -d saveetha.com -b all -r -f saveetha_resolved (Scan 2: the same sources with -r to resolve each host name to its IP address; saved to saveetha_resolved.json / .xml / .jsonl).

Sources that need an API key (for example Censys, Hunter, Shodan, VirusTotal, SecurityTrails) were skipped by the tool with the message "missing credentials". The free sources (for example crt.sh, Certspotter, RapidDNS, Robtex, URLScan, AlienVault OTX, HackerTarget, Wayback Archive and others) ran normally.

Fill the following required details

Required detail	Result
Domain Name	saveetha.com
Emails found (List all one by one)	None found. theHarvester reported "No emails found." in both scans (0 e-mail addresses).
Sub emails found (List all one by one)	None found (0). No people or sub e-mail results were returned either ("No people found.").
Hosts and Its IP (List all one by one)	All 129 hosts with their IP addresses are listed one by one in Table 1 below. 99 hosts resolved to at least one IP address; 30 hosts were discovered but returned no IP in the saved results.
Total Email Addresses Found	0
Total Subdomains Found	129 (unique host names under saveetha.com; 36 of them begin with "www.")
Total Hosts Found	129
Total IP Addresses Found	36 unique addresses (35 IPv4 and 1 IPv6). The 36 addresses are listed in Table 2.

Screenshot of Command execution	See Section “Evidence: Command Execution” (terminal capture of the commands and the tool banner).
Screenshot of theHarvester output	See Section “Evidence: theHarvester Output” (terminal capture of the results).

Note: the tool’s on-screen summary lines printed larger numbers (for example “IPs found: 54” in Scan 2 and “23” in Scan 1) than the number of addresses it actually listed and saved. The totals in this report are counted from the addresses and host names listed in the saved JSON files, which can be checked item by item.

Table 1: Hosts and Their IP Addresses (all 129, one by one)

No.	Host name	IP address(es)
1	360.saveetha.com	15.252.235.72, 15.252.80.197, 3.109.232.78
2	admission.saveetha.com	43.204.251.6
3	admission.sse.saveetha.com	43.204.251.6
4	admissions.saveetha.com	43.204.251.6
5	anat.sdc.saveetha.com	198.185.159.144, 198.185.159.145, 198.49.23.144, 198.49.23.145
6	api.vbuy.saveetha.com	No IP returned in saved results
7	arms.libarts.saveetha.com	103.249.82.249
8	arms.scad.saveetha.com	103.249.82.245
9	arms.scahs.saveetha.com	103.249.82.249
10	arms.scon.saveetha.com	103.249.82.249
11	arms.scop.saveetha.com	103.249.82.245
12	arms.scpt.saveetha.com	103.249.82.245
13	arms.sdc.saveetha.com	106.51.116.185
14	arms.shost.saveetha.com	103.249.82.249
15	arms.sibms.saveetha.com	103.249.82.249
16	arms.smc.saveetha.com	103.249.82.249
17	arms.soct.saveetha.com	103.249.82.249
18	arms.sse.saveetha.com	103.249.82.245
19	arms.ssl.saveetha.com	106.51.116.185
20	arms.ssm.saveetha.com	106.51.116.185
21	arms.sspe.saveetha.com	103.249.82.245
22	biochem.sdc.saveetha.com	198.185.159.144, 198.185.159.145, 198.49.23.144, 198.49.23.145
23	brulac.sdc.saveetha.com	198.185.159.144
24	civil.sse.saveetha.com	172.217.31.243, 2404:6800:4009:821::2013
25	cse.sse.saveetha.com	14.139.187.237
26	dias.sdc.saveetha.com	182.19.38.181
27	diasfiles.saveetha.com	182.19.38.181
28	elearning.dental.saveetha.com	No IP returned in saved results
29	empowher.saveetha.com	180.235.121.249
30	endo.sdc.saveetha.com	198.185.159.144, 198.185.159.145, 198.49.23.144, 198.49.23.145
31	epay.saveetha.com	No IP returned in saved results
32	esthetics.sdc.saveetha.com	198.185.159.144, 198.185.159.145, 198.49.23.144, 198.49.23.145
33	food.saveetha.com	No IP returned in saved results
34	gpath.sdc.saveetha.com	198.185.159.144, 198.185.159.145, 198.49.23.144, 198.49.23.145
35	hostel.saveetha.com	No IP returned in saved results
36	implant.sdc.saveetha.com	No IP returned in saved results
37	isss.saveetha.com	No IP returned in saved results
38	libarts.saveetha.com	199.34.228.77
39	life.saveetha.com	No IP returned in saved results
40	lms.sse.saveetha.com	No IP returned in saved results
41	mail.saveetha.com	No IP returned in saved results
42	mias.files.smc.saveetha.com	103.249.82.253
43	mias.smc.saveetha.com	103.249.82.253
44	micro.sdc.saveetha.com	198.185.159.144, 198.185.159.145, 198.49.23.144, 198.49.23.145
45	naac.saveetha.com	198.185.159.144, 198.185.159.145, 198.49.23.144, 198.49.23.145

No.	Host name	IP address(es)
46	nmc.mias.smc.saveetha.com	103.249.82.253
47	omed.sdc.saveetha.com	198.185.159.144, 198.185.159.145, 198.49.23.144, 198.49.23.145
48	omfs.sdc.saveetha.com	198.185.159.144, 198.185.159.145, 198.49.23.144, 198.49.23.145
49	online.saveetha.com	3.110.122.44
50	opath.sdc.saveetha.com	198.185.159.144, 198.185.159.145, 198.49.23.144, 198.49.23.145
51	oralcancerinstitute.sdc.saveetha.com	198.185.159.144, 198.185.159.145, 198.49.23.144, 198.49.23.145
52	ortho.sdc.saveetha.com	198.185.159.144, 198.185.159.145, 198.49.23.144, 198.49.23.145
53	pacs.saveetha.com	180.235.121.243
54	pacs.vhealth.saveetha.com	No IP returned in saved results
55	pedo.sdc.saveetha.com	198.185.159.144, 198.185.159.145, 198.49.23.144, 198.49.23.145
56	perio.sdc.saveetha.com	198.185.159.144, 198.185.159.145, 198.49.23.144, 198.49.23.145
57	pharm.sdc.saveetha.com	198.185.159.144, 198.185.159.145, 198.49.23.144, 198.49.23.145
58	phd.sdc.saveetha.com	198.185.159.144, 198.185.159.145, 198.49.23.144, 198.49.23.145
59	physio.sdc.saveetha.com	198.185.159.144, 198.185.159.145, 198.49.23.144, 198.49.23.145
60	portal.saveetha.com	14.139.187.227
61	prime.sdc.saveetha.com	13.233.50.50
62	prostho.sdc.saveetha.com	199.34.228.77
63	said.sdc.saveetha.com	198.49.23.145
64	scad.saveetha.com	198.185.159.144, 198.185.159.145
65	scahs.saveetha.com	34.149.87.45
66	scet.saveetha.com	No IP returned in saved results
67	scon.saveetha.com	198.49.23.144
68	scot.saveetha.com	34.149.87.45
69	scp.saveetha.com	34.149.87.45
70	scpt.saveetha.com	198.49.23.145
71	sdc.saveetha.com	198.185.159.144
72	sdg.saveetha.com	180.235.121.250
73	simatsadmissions.saveetha.com	43.204.251.6
74	simatsentrance.saveetha.com	43.204.251.6
75	simatsexam.saveetha.com	172.21.100.231
76	simatslab.saveetha.com	180.235.121.254
77	simatstransport.saveetha.com	180.235.121.244
78	smartcard.saveetha.com	No IP returned in saved results
79	smc.saveetha.com	3.110.122.44
80	sse.saveetha.com	198.185.159.144
81	ssl.saveetha.com	198.185.159.144, 198.185.159.145, 198.49.23.144, 198.49.23.145
82	ssm.saveetha.com	198.49.23.145
83	sspe.saveetha.com	198.185.159.144, 198.185.159.145, 198.49.23.144, 198.49.23.145
84	thepupil.saveetha.com	14.139.187.237
85	vbuy.saveetha.com	13.204.6.30
86	vconnect.saveetha.com	13.233.98.211
87	vhealth.saveetha.com	13.235.108.75
88	vquest.saveetha.com	103.249.82.246
89	vstay.saveetha.com	15.206.172.50
90	vstudy.saveetha.com	43.204.251.6
91	vtrip.saveetha.com	43.204.251.6
92	webmail.saveetha.com	No IP returned in saved results
93	whitelab.sdc.saveetha.com	198.49.23.144
94	www.arms.shost.saveetha.com	No IP returned in saved results
95	www.arms.sibms.saveetha.com	No IP returned in saved results
96	www.arms.sse.saveetha.com	No IP returned in saved results
97	www.brulac.sdc.saveetha.com	198.185.159.144, 198.185.159.145, 198.49.23.144, 198.49.23.145
98	www.endo.sdc.saveetha.com	No IP returned in saved results
99	www.esthetics.sdc.saveetha.com	No IP returned in saved results
100	www.gpath.sdc.saveetha.com	No IP returned in saved results
101	www.implant.sdc.saveetha.com	No IP returned in saved results
102	www.isss.saveetha.com	No IP returned in saved results

No.	Host name	IP address(es)
103	www.lms.smc.saveetha.com	14.98.52.3
104	www.mias.files.smc.saveetha.com	No IP returned in saved results
105	www.mias.smc.saveetha.com	No IP returned in saved results
106	www.naac.saveetha.com	198.185.159.144, 198.185.159.145, 198.49.23.144, 198.49.23.145
107	www.nmc.mias.smc.saveetha.com	No IP returned in saved results
108	www.omed.sdc.saveetha.com	No IP returned in saved results
109	www.oralcancerinstitute.sdc.saveetha.com	No IP returned in saved results
110	www.phd.sdc.saveetha.com	No IP returned in saved results
111	www.physio.sdc.saveetha.com	No IP returned in saved results
112	www.portal.saveetha.com	14.139.187.227
113	www.said.sdc.saveetha.com	198.185.159.144, 198.185.159.145, 198.49.23.144, 198.49.23.145
114	www.saveetha.com	198.185.159.144
115	www.scad.saveetha.com	198.185.159.144, 198.185.159.145, 198.49.23.144, 198.49.23.145
116	www.scahs.saveetha.com	34.149.87.45
117	www.scon.saveetha.com	198.185.159.144, 198.185.159.145, 198.49.23.144, 198.49.23.145
118	www.scot.saveetha.com	34.149.87.45
119	www.scp.saveetha.com	34.149.87.45
120	www.scpt.saveetha.com	198.185.159.144, 198.185.159.145, 198.49.23.144, 198.49.23.145
121	www.sdc.saveetha.com	198.185.159.144, 198.185.159.145, 198.49.23.144, 198.49.23.145
122	www.simatstransport.saveetha.com	No IP returned in saved results
123	www.smc.saveetha.com	3.110.122.44
124	www.sse.saveetha.com	198.185.159.144, 198.185.159.145, 198.49.23.144, 198.49.23.145
125	www.ssl.saveetha.com	198.185.159.144, 198.185.159.145, 198.49.23.144, 198.49.23.145
126	www.ssm.saveetha.com	198.185.159.144, 198.185.159.145, 198.49.23.144, 198.49.23.145
127	www.sspe.saveetha.com	198.185.159.144, 198.185.159.145, 198.49.23.144, 198.49.23.145
128	www.vstudy.saveetha.com	43.204.251.6
129	www.whitelab.sdc.saveetha.com	198.185.159.144, 198.185.159.145, 198.49.23.144, 198.49.23.145

Source: saveetha_results.json (host list, Scan 1) and saveetha_resolved.json (host-to-IP pairs, Scan 2). Hosts with several addresses list all of them.

Table 2: IP Addresses Found (all 36, one by one)

No.	IP address	No. of hosts	Hosts using it (names shortened, .saveetha.com omitted)	Remarks
1	3.109.232.78	1	360	Public IPv4 address; no ownership reported by the tool
2	3.110.122.44	3	online, smc, www.smc	Tool attribution: AS16509 AMAZON-02 (Amazon.com, Inc., US)
3	13.204.6.30	1	vbuy	Public IPv4 address; no ownership reported by the tool
4	13.233.50.50	1	prime.sdc	Public IPv4 address; no ownership reported by the tool
5	13.233.98.211	1	vconnect	Public IPv4 address; no ownership reported by the tool
6	13.235.108.75	1	vhealth	Public IPv4 address; no ownership reported by the tool
7	14.98.51.2	0	None in the saved host-to-IP pairs	Public IPv4 address reported by theHarvester in the IP list; no ownership reported by the tool
8	14.98.52.3	1	www.lms.smc	Public IPv4 address; no ownership reported by the tool
9	14.139.187.227	2	portal, www.portal	Public IPv4 address; no ownership reported by the tool
10	14.139.187.237	2	cse.sse, thepupil	Public IPv4 address; no ownership reported by the tool
11	15.206.172.50	1	vstay	Public IPv4 address; no ownership reported by the tool
12	15.252.80.197	1	360	Public IPv4 address; no ownership reported by the tool

No.	IP address	No. of hosts	Hosts using it (names shortened, .saveetha.com omitted)	Remarks
13	15.252.235.72	1	360	Public IPv4 address; no ownership reported by the tool
14	34.149.87.45	6	scahs, scot, scp and 3 more	Public IPv4 address; no ownership reported by the tool
15	43.204.251.6	8	admission, admission.sse, admissions and 5 more	Public IPv4 address; no ownership reported by the tool
16	43.205.1.154	0	None in the saved host-to-IP pairs	Public IPv4 address reported by theHarvester in the IP list; no ownership reported by the tool
17	103.249.82.245	5	arms.scad, arms.scop, arms.scpt and 2 more	Public IPv4 address; no ownership reported by the tool
18	103.249.82.246	1	vquest	Public IPv4 address; no ownership reported by the tool
19	103.249.82.249	7	arms.libarts, arms.scahs, arms.scon and 4 more	Public IPv4 address; no ownership reported by the tool
20	103.249.82.253	3	mias.files.smc, mias.smc, nmc.mias.smc	Public IPv4 address; no ownership reported by the tool
21	106.51.116.185	3	arms.sdc, arms.ssl, arms.ssm	Public IPv4 address; no ownership reported by the tool
22	118.139.175.1	0	None in the saved host-to-IP pairs	Public IPv4 address reported by theHarvester in the IP list; no ownership reported by the tool
23	172.21.100.231	1	simatsexam	Private (RFC 1918) address, not routable on the internet; found for simatsexam.saveetha.com
24	172.217.31.243	1	civil.sse	Public IPv4 address; no ownership reported by the tool
25	180.235.121.243	1	pacs	Public IPv4 address; no ownership reported by the tool
26	180.235.121.244	1	simatstransport	Public IPv4 address; no ownership reported by the tool
27	180.235.121.249	1	empowher	Public IPv4 address; no ownership reported by the tool
28	180.235.121.250	1	sdg	Public IPv4 address; no ownership reported by the tool
29	180.235.121.254	1	simatslab	Public IPv4 address; no ownership reported by the tool
30	182.19.38.181	2	dias.sdc, diasfiles	Public IPv4 address; no ownership reported by the tool
31	198.49.23.144	33	anat.sdc, biochem.sdc, endo.sdc and 30 more	Public IPv4 address; no ownership reported by the tool
32	198.49.23.145	34	anat.sdc, biochem.sdc, endo.sdc and 31 more	Public IPv4 address; no ownership reported by the tool
33	198.185.159.144	36	anat.sdc, biochem.sdc, brulac.sdc and 33 more	Tool attribution: AS53831 SQUARESPACE (Squarespace, Inc., US)
34	198.185.159.145	32	anat.sdc, biochem.sdc, endo.sdc and 29 more	Public IPv4 address; no ownership reported by the tool
35	199.34.228.77	2	libarts, prostho.sdc	Public IPv4 address; no ownership reported by the tool
36	2404:6800:4009:821::2013	1	civil.sse	IPv6 address; found for civil.sse.saveetha.com

"No. of hosts" is counted from the host-to-IP pairs of Scan 2. Every IP listed here appears in the saved IP lists of the tool; the remarks use only what theHarvester itself reported (ASN attributions) or what the address format shows (private or IPv6).

Table 3: Other Information Reported (Scan 1)

Item	Value reported by theHarvester
ASNs found (2)	AS16509, AS53831

Item	Value reported by theHarvester
ASN attributions (3)	AS16509 AMAZON-02 – Amazon.com, Inc., US (IP 3.110.122.44); AS53831 SQUARESPACE – Squarespace, Inc., US (host www.saveetha.com and IP 198.185.159.144)
URLs found (2)	http://saveetha.com/ ; https://www.saveetha.com/
Emails / People	No emails found; No people found

Evidence: Command Execution

Terminal text copied from the Windows command prompt (C:\Users\ANJI\theHarvester). The first block shows the tool installed and verified; the second shows the scan command being run.

```
C:\Users\ANJI>pip install theHarvester
Successfully installed theHarvester-0.0.1
C:\Users\ANJI>theHarvester --help
'theHarvester' is not recognized as an internal or external command,


C:\Users\ANJI>git clone https://github.com/laramies/theHarvester.git
C:\Users\ANJI>cd theHarvester
C:\Users\ANJI\theHarvester>uv sync
C:\Users\ANJI\theHarvester>uv run theHarvester -h
* theHarvester 5.0.0
* Coded by Christian Martorella
usage: theHarvester [-h] -d DOMAIN [-l LIMIT] ... [-f NAME] ... [-b SOURCE]
```

```
C:\Users\ANJI\theHarvester>uv run theHarvester -d saveetha.com -b all -r -f saveetha_resolved
* theHarvester 5.0.0
[*] Activity: P0 passive collection, P1 DNS interaction
[*] Target: saveetha.com
[*] Source workers: requested=3; effective=3.
[*] Searching Apis-guru.
[*] Searching Arquivo.
[*] Searching Baidu.
[!] Source bevigil skipped: missing credentials.
[*] Searching Certspotter.
[*] Searching Crtsh.
[*] Searching Hackertarget.
[*] Searching Rapiddns.
[*] Searching Urlscan.
   ... (remaining sources)
[*] Reporting started.
[*] XML File saved.
[*] JSON File saved.
[*] JSONL File saved.
```

Evidence: theHarvester Output

Result lines from the terminal (Scan 1, then Scan 2 with host-to-IP resolution). Full lists are in Tables 1 to 3.

```
[*] ASNS found: 2
AS16509
AS53831
[*] ASN organization attributions found: 3
AS16509 | AMAZON-02 - Amazon.com, Inc., US | source:urlscan | ip:3.110.122.44
AS53831 | SQUARESPACE - Squarespace, Inc., US | source:urlscan | hostname:www.saveetha.com
AS53831 | SQUARESPACE - Squarespace, Inc., US | source:urlscan | ip:198.185.159.144
[*] URLs found: 2
http://saveetha.com/
https://www.saveetha.com/
[*] No emails found.
[*] No people found.
[*] Hosts found: 129
360.saveetha.com
admission.saveetha.com
admission.sse.saveetha.com
...
[*] Reporting started.  [*] XML File saved.  [*] JSON File saved.  [*] JSONL File saved.
```

```
--- Scan 2 (-r) ---
360.saveetha.com:15.252.235.72,15.252.80.197,3.109.232.78
admission.saveetha.com:43.204.251.6
anat.sdc.saveetha.com:198.185.159.144,198.185.159.145,198.49.23.144,198.49.23.145
arms.sdc.saveetha.com:106.51.116.185
simatsexam.saveetha.com:172.21.100.231
www.saveetha.com:198.185.159.144
...
```

Saved result files: saveetha_results.json, saveetha_results.xml, saveetha_results.jsonl, saveetha_resolved.json, saveetha_resolved.xml and saveetha_resolved.jsonl (folder C:\Users\ANJI\theHarvester).

Analysis and Interpretation of Results

1. Overview

theHarvester collected passive (public-source) information about saveetha.com and found 129 host names, 36 IP addresses, 2 ASNs and 2 URLs, but no e-mail addresses. All 129 hosts are subdomains of saveetha.com, which shows that the domain is the root of a large web presence made up of many separate sites and services rather than a single website.

2. Emails (0 found)

No e-mail addresses were found. This does not prove that the organisation has no addresses. The sources that usually return e-mail addresses (for example Hunter, Censys, Intelx, RocketReach and Tomba) need API keys and were skipped, and the free search-engine sources return only what happens to be indexed publicly. Staff addresses on university domains are often hidden from search engines or published as images. From a defender's point of view, an empty result is a good sign because it means addresses are not easily harvested, which reduces the material available for phishing and social engineering against the domain.

3. Subdomains and hosts (129)

The host names show the structure of the organisation. 36 hosts sit under sdc.saveetha.com (for example anat.sdc, endo.sdc, ortho.sdc, perio.sdc, pharm.sdc), which are departmental sites of one institution. 18 hosts begin with arms. or www.arms. (arms.scad, arms.sdc, arms.smc, arms.sse and others), which suggests a common web application deployed once per institute. Further groups are the "v" services (vbuy, vconnect, vhealth, vquest, vstay, vstudy, vtrip), administrative and student services (admission, admissions, epay, hostel, isss, portal, smartcard, simatsexam, simatsentrance, simatsadmissions, simatslab, simatstransport) and mail services (mail, webmail). The 36 www. entries are alias names of other hosts. These observations are drawn from the host names only; the tool does not state what each system does.

4. IP addresses and hosting (36)

- **Third-party website platform.** Squarespace-hosted sites: the tool attributes 198.185.159.144 to AS53831 (Squarespace, Inc., US). In Scan 2, 198.185.159.144, 198.185.159.145, 198.49.23.144 and 198.49.23.145 are the most common addresses, with 36, 32, 33 and 34 hosts respectively (including www.saveetha.com, sdc, naac and ssl). The 198.49.23.x addresses were not attributed by the tool, but they are consistent with the same hosted-website platform. This

means the public websites of many institutes are served by a third-party website platform, not by the organisation's own servers.

- **Cloud hosting.** Addresses such as 3.110.122.44 (AS16509 AMAZON-02, attributed by the tool; hosts online, smc and www.smc), 43.204.251.6 (8 hosts: admission, admissions, simatsadmissions, simatsentrance, vstudy, vtrip and others) and the 13.x, 15.x and 3.x addresses used by the "v" services and 360.saveetha.com point to cloud hosting. Only 3.110.122.44 was attributed to Amazon by the tool itself.
- **Probable self-hosted systems.** Addresses in the 103.249.82.x range (7 hosts on .249, 5 on .245, 3 on .253, 1 on .246) host the arms.* and mias.* hosts, and addresses in the 180.235.121.x and 14.139.187.x ranges host hosts such as pacs, portal, simatslab, simatstransport and thepupil. These look like the institution's own or its internet provider's address space, which would mean part of the infrastructure is self-hosted. The tool did not attribute these ranges, so this is an interpretation, not a stated fact.
- **Other.** civil.sse.saveetha.com resolves to 172.217.31.243 and the IPv6 address 2404:6800:4009:821::2013, showing that the site also uses IPv6. 34.149.87.45 is used by scahs, scot and scp (and their www. names).

5. Security significance

- **Large attack surface.** The 129 host names give an attacker a map of the organisation. Each one is a separate system that needs patching, access control and monitoring. Staging, exam, laboratory, payment and medical-imaging names (simatsexam, simatslab, epay, pacs) are higher-value targets than a brochure website.
- **Private address in public DNS.** simatsexam.saveetha.com resolves to 172.21.100.231, a private (RFC 1918) address. A public DNS record that points to an internal address reveals internal network addressing and can indicate a record that was left in public DNS by mistake. It is not reachable from the internet, so it is an information leak, not an exploitable hole.
- **Hosts without an IP.** Four hosts were discovered but did not resolve to any IP address in Scan 2 (elearning.dental, epay, lms.sse and mail). Records that exist but point nowhere should be reviewed, because forgotten records pointing at external services can sometimes be claimed by someone else (subdomain takeover). No takeover check was run (the -t option was not used), so this is only a reason for review and not a finding.
- **Mixed hosting.** Hosts spread over cloud, website-platform and self-hosted networks need separate security settings on each platform. Using a managed platform for brochure sites is good practice, as long as the self-hosted application servers (arms.*, pacs, portal) are kept patched and restricted.
- **Passive only.** All data was collected passively from public sources; no host was scanned, logged in to or exploited. The findings are indications from public DNS and certificate data, not confirmed vulnerabilities.

6. Limitations

The results depend on the sources that were available and on those that did not need API keys. Several sources (Censys, Hunter, Shodan, VirusTotal, SecurityTrails and others) were skipped, one Chromium-based search source failed to start on Windows, and the Common Crawl source returned errors. The two scans did not return exactly the same lists: Scan 2 resolved 103 host names, of which 99 had an IP address, while Scan 1 listed 129 host names, and every host found in Scan 2

also appears in Scan 1. A scan with API keys would probably find more hosts and might find e-mail addresses.

Conclusion

Domain and e-mail footprinting of saveetha.com with theHarvester 5.0.0 found 0 e-mail addresses, 129 hosts (all subdomains) and 36 IP addresses, spread mainly over a third-party website platform (Squarespace, AS53831), cloud hosting (Amazon, AS16509, and other cloud-looking ranges) and addresses that appear to be the institution's own or its provider's space. The results show how much of an organisation's infrastructure can be learned from public sources without touching its systems, and they point to three items for review: the private address published for simatsexam.saveetha.com, the four hosts that did not resolve, and the large number of separately managed hosts.